

48 Neurons, Synapses, and Signaling

KEY CONCEPTS

- 48.1** Neuron structure and organization reflect function in information transfer p. 1068
- 48.2** Ion pumps and ion channels establish the resting potential of a neuron p. 1069
- 48.3** Action potentials are the signals conducted by axons p. 1072
- 48.4** Neurons communicate with other cells at synapses p. 1077

Study Tip

Make a table: The term *potential* is central to understanding neuronal signaling, as it provides the driving force for both electrical and chemical signaling. Fill in a table like this one as you read to help you distinguish characteristics of a *resting potential*, *graded potential*, and *action potential*, as well as chemical potentials for sodium and potassium ion concentrations.

Membrane potentials of a neuron		Occurs in cell body?	Occurs in axon?	Variable strength?
Electrical	Resting	Yes	Yes	No
Electrical	Graded			
Electrical	Action			
Chemical	[Na ⁺]			
Chemical	[K ⁺]			

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 48
- BioFlix® Animation: How Neurons Work
- BioFlix® Animation: How Synapses Work

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Interpreting Data Values Expressed in Scientific Notation

Ready-to-Go Teaching Module

(in Instructor Resources)

- Resting and Action Potentials



Figure 48.1 This cone snail (*Conus geographus*) is slow-moving, but a dangerous hunter. Delivering venom with a hollow, harpoon-like tooth, it can paralyze a fish almost instantaneously. Scuba divers picking up a cone snail have died from a single injection. Underlying the fast-acting, lethal nature of the venom is an ability to block information transfer by neurons, specialized cells of the nervous system.

How does a neuron transmit information?

A neuron receives information, transmits it along an extension called an **axon**, and transmits the information to other cells via specialized junctions called **synapses**.

